Multi-task Learning with Adaptive Global Temporal Structure for Predicting Alzheimer's Disease Progression

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ABSTRACT

In this paper, we propose a multi-task learning approach for predicting the progression of Alzheimer's disease (AD), known as the most common form of dementia. The vital challenge is to identify how the tasks are related and build learning models to capture such task relatedness. Unlike previous methods that assume low-rank structure, chase the predefined local temporal relatedness or utilize local approximation, we propose a novel penalty termed **L**ongitudinal **S**tability **A**djustment (LSA) to adaptively capture the intrinsic global temporal correlation among multiple time points and thus utilize the accumulated disease progression information. We combine LSA with sparse group Lasso to present a novel multi-task learning formulation to identify biomarkers closely related to cognitive measurement and predict AD progression. Two efficient algorithms are designed for large-scale dataset. Experimental results conducted on two AD data sets demonstrate our framework outperforms competing methods in terms of overall and each task performances. We also perform stability selection to identify stable biomarkers from the MRI feature set and analyze their temporal patterns in disease progression.

CCS CONCEPTS

• **Computing methodologies** → **Multi-task learning**; • **Information systems** → **Data mining**.

KEYWORDS

Alzheimer's Disease, disease progression, multi-task learning, adaptive global temporal relation, identification of biomarkers

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1 INTRODUCTION

Alzheimer's disease (AD) is known as the most common cause of dementia, directly leading to death, which accounts for an estimated 60% to 80% of cases [2]. Total payments in 2021 in the U.S. for all Americans with AD or other dementias are estimated at $355 billion, not including the value of informal caregiving [37]. There is no cure for AD, it is hence significant to study biomarkers that play an important role in the development of AD [2]. On the other hand, treatment is more effective in the early stage of AD, however, the definite diagnosis of AD can only be made through a complicated brain biopsy or autopsy [45], so for AD research, there are two urgent needs:

- Accurately predict disease progression over time.
- Identify the most informative biomarkers.

Previous studies have shown that various cognitive scores, such as Mini Mental State Examination (MMSE) and the Alzheimer's Disease Assessment Scale Cognitive Subscale (ADAS), can be used to estimate the state of AD patients and MRI biomarkers closely related to AD [1]. Advanced machine learning techniques have been utilized to model MRI features and cognitive scores to achieve the two objectives mentioned above. Related learning problems usually include classification, survival analysis, and regression models.

Classification-based models [22] aim to classify the patient's state into a series of predefined disease stages. Survival analysis models [34] can answer some questions like how long the patient can live. However, neither of these two methods above can accurately predict the patient's disease state. Deep learning approaches [21] may be limited by the interpretability of the model, which does not always allow to perform the pattern localization [23]. Regression methods of single-task learning usually do not consider the cognitive state of patients at multiple time points in the future. Since the underlying pathology of the disease is the same over time,

the cognitive scores of the same cognitive measure at different time points have obvious inherent correlation. Therefore, a joint analysis of tasks at multiple time points via multi-task learning is expected to improve the performance, especially in the case the feature number is fairly larger than the sample number.

The idea of multi-task learning [6] is to utilize the intrinsic relationships among multiple related tasks to improve the generalization performance. Although great progress in predicting disease progression via multi-task learning has been made during the past few years, the vital limitation is to identify how the tasks are related and build learning models to capture such task relatedness. Motivated by an intuitive idea that the practical diagnosis of disease should not only rely on current symptoms of patient; but also consider their previous states, we propose a novel multi-task learning approach with *global temporal relatedness* among tasks. To be specific, we first define the disease progression value at kth time point as $\delta y_k = y_{k+1} - y_k$, y_k is the patient status at kth time point, usually measured by some quantitative cognitive scores, such as MMSE and ADAS. We assume the current time point δy_k is relevant to all previous progression values, $\delta y_i, i = [1, \cdots, k-1]$. Due to the utilization of square loss, this global temporal relatedness of the cognitive score is equivalent to the relationship between tasks, and explains why this assumption works: the influence of biomarkers like cerebrospinal fluid [14] is not identical over time and the importance of biomarkers at a current time point is related to previous time points. One iterative matrix multiplication method termed **L**ongitudinal **S**tability **A**djustment (LSA) is utilized to model this idea such that the relatedness matrix among tasks (time points) can be determined adaptively, meaning the accumulative progression information is incorporated into our approach. The sparse group Lasso penalty [32] is applied to select a common feature set for for all tasks and a specific set of features for each task.

Note that, although only AD data set are considered in this paper due to the limitation of paper space, our framework is general for modeling the progression of various diseases, such as Parkinson's [10] and diabetes [31].

The proposed formulations are challenging to solve due to the utilization of three nonsmooth penalties as well for dealing with large-scale AD data sets, we first design an optimization algorithm based on the accelerated gradient method (AGM) which has the optimal convergence rate among first-order methods[27]. However, on account of no analytical solution of proximal operator for those penalties, the optimal convergence rate can not be preserved in a relatively high accuracy case[16]. So we another design an efficient optimization algorithm based on inexact the well-developed alternating direction method of multipliers (inexact ADMM) [15] which is shown to have the same convergence rate as the exact strategy. We perform extensive experiments on the newest dataset we have processed from the Alzheimer's Disease Neuroimaging Initiative (ADNI). Through empirical evaluation, we illustrate our framework outperforms competing methods with the best prediction performance and robustness.

We summarize our contributions as follows:

- The novel approach LSA we have proposed is capable of adaptively capturing the global temporal task relation in AD progression. This adaptive global temporal structure will take into account the overall progression of the disease, thus helps our proposed multi-task framework MTL-LSA be more accurate and robust than multiple baseline methods.
- The AGM-based optimization algorithm is designed for processing large-scale AD data. To solve the problem that the superlinear convergence rate of AGM is no longer in the high precision scenario, we design another algorithm based on inexact ADMM, which decouples the model coefficient matrix W by suitably linearization method to greatly improve the efficiency.
- To further show the high interpretability of our formulations, which existing deep learning methods [21][9] [12][39] [38][42] fail to have, we also identify and analyze the temporal pattern of stable MRI biomarkers. Where the results are clinically meaningful and have a potential impact on identifying the most predictive biomarkers for AD.

2 RELATED WORK

In [46], multi-task learning termed TGL is used to model the longitudinal disease progression. TGL constrains the models at all time points to select a common set of features, thus fails to capture the temporal patterns of the biomarkers [19]. To address this problem, the sparse group Lasso penalty[32] is applied in the new MTL framework cFSGL [44] to conduct joint feature selection for all tasks and specific feature selection for each task.

The vital idea of TGL and cFSGL is the temporal smoothness assumption means the difference of cognitive scores between two successive time points is relatively small, i.e., chases the local smoothness over time. In [31], a spatio-temporal approach using Laplacian-based penalty is proposed for diabetes-related complication. In [35], a tensor-based temporal survival approach is presented for predicting the time to event of interest along with its probability on longitudinal data. The core idea of above two approaches is still the local temporal assumption. However, this assumption does not consider the global temporal connection between tasks, potentially missing out on helpful task dependencies beyond the immediate neighbors. In [25], they consider weighted task relation matrix based on a Gaussian kernel over time points, yielding a fully connected graph. The possible limitation is this kind of local approximation can not fully exploit the complex correlation among multiple time points.

Some well-developed MTL frameworks are proposed recently. DMTL [20] is designed for modeling dirty data which may not fall into a single neat structural bracket (all block-sparse, all low-rank and so on). A robust MTL algorithm [8] is proposed to learn multiple tasks simultaneously as well as identify the irrelevant (outlier) tasks. RMTFL [13] simultaneously captures a common set of features among relevant tasks and identifies outlier tasks. A nonconvex calibrated multi-task learning [28] adopts a nonconvex low rank regularizer to explore the shared information among different tasks. Although, these algorithms have great performance on multi data sets, but not specifically designed for modeling disease progression and thus fail to take full advantage of the complicated correlation among tasks.

Different from existing methods, our proposed LSA adaptively capture the global temporal relation among tasks. Then based on

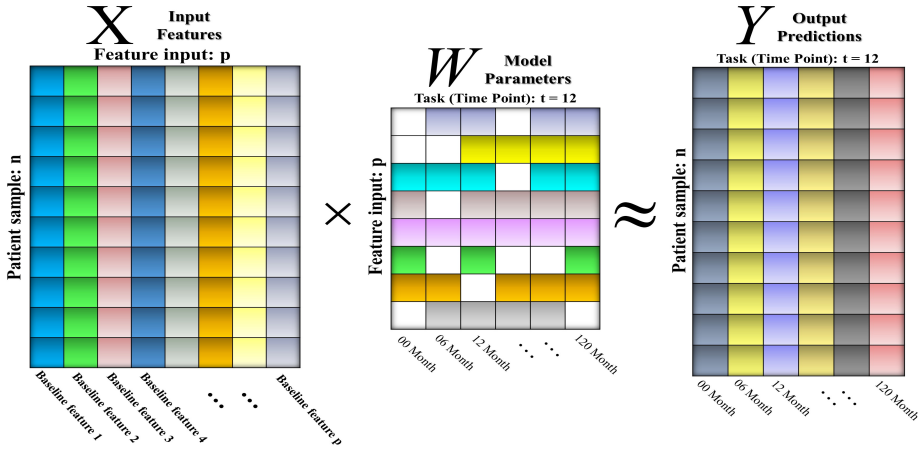

Figure 1: Illustration of a simple special case of MTL prediction model, in which samples for all tasks are same, i.e., $X_i, i \in \{1, \cdots, t\}$ **are identical. Specifically, each row of** X_i **represents all baseline MRI features of one patient for** i**-th task. Each column of** X_i **corresponds to a kind of feature at baseline.** $W = [\boldsymbol{w_1}, \cdots, \boldsymbol{w_t}] \in R^{p \times t}$ **as the weight matrix. In the prediction model, we assume a linear relationship between** X **and** Y**.**

LSA, the framework MTL-LSA achieves the best prediction performance and are more robust than comparative models. In the meantime, our framework preserves the capability to identify the most significant and informative biomarkers, regarded as high interpretability, while deep learning models [21, 38, 39] fail to have.

3 METHODS

3.1 Multi-task Learning

Consider we have a multi-task learning problem with t tasks, each task $i \in \{1, \cdots, t\}$ is associated with a set of samples $(X_i, \boldsymbol{y_i}), X_i \in R^{n_i \times p}, \boldsymbol{y_i} \in R^{n_i}$. We denote $X = [X_1, \cdots, X_t], Y = [\boldsymbol{y_1}, \cdots, \boldsymbol{y_t}]$ for simple representation. Let $W = [\boldsymbol{w_1}, \cdots, \boldsymbol{w_t}] \in R^{p \times t}$ denote the regression parameter matrix over all tasks. In the context of AD, each time point corresponds to one single task. Concretely, each row of X_i represents all features of one patient for ith task. Each column of X_i represents a kind of MRI biomarker (feature) at the baseline time point. For the output matrix $Y = [\boldsymbol{y_1}, \cdots, \boldsymbol{y_t}]$, each column corresponds to the score at a future time point as well as task. We assume the loss to be square loss: $L(Y, X, W) = \frac{1}{2}\sum_{i=1}^{t} \|X_i \boldsymbol{w_i} - \boldsymbol{y_i}\|_2^2 = \frac{1}{2}\|XW - Y\|_F^2$. Fig. 1 is the illustration of a special case of every task has same instance data. The input data is the MRI feature set at start time point (baseline), the notation "x Month" indicates x months after baseline time point.

3.2 Longitudinal Stability Adjustment

We first define the ith progression of AD for one patient, denoted as δy_i, as: $\delta y_i = y_i - y_{i+1}, i = \{1, 2, \cdots, t-1\}$, y_i represents the cognitive score of one specific patient at time point i (task i in multi-task framework). It is manifest the kth progression δy_k is related to all previous progressions $\delta y_i, i = 1, \cdots, k-1$. We give a new definition, named "true progression", denoted as Δy.

Definition 3.1. *The ith true progression of disease Δy_i satisfies*

$$\begin{cases} \Delta y_1 = \delta y_1 \\ \Delta y_2 = \alpha \Delta y_1 + (1-\alpha)\delta y_2 \\ \cdots \\ \Delta y_{t-1} = \alpha \Delta y_{t-2} + (1-\alpha)\delta y_{t-1}. \end{cases} \tag{1}$$

In the above Definition 3.1, the parameter $\alpha \in [0, 1]$ represents the relational degree of the ith progression and all previous progressions. Obviously the disease progression information accumulated over time is integrated by this sort of iterative convex combination method. Now according to (1), after denoting $\boldsymbol{y} = [y_1, \cdots, y_t]$, we formulate LSA via the following matrix multiplication:

$$\boldsymbol{y}H(\alpha) = \boldsymbol{y}\Theta A_1(\alpha)A_2(\alpha)\cdots A_{t-2}(\alpha), \tag{2}$$

where the matrix $\Theta \in R^{t \times (t-1)}$ satisfies $\Theta_{ij} = 1$ if $i = j, \Theta_{ij} = -1$ if $i = j+1$, and $\Theta_{ij} = 0$ otherwise. The matrix $H(\alpha)$, representing the correlation among progression, is a function of the parameter α, which depends on the result of cross validation in our experimental setting. $A_i(\alpha) \in R^{(t-1)\times(t-1)}$ is an identity matrix, and the value of $A_{i_{m,n}}(\alpha)$ is replaced by α if $m = i, n = i+1$, the value of $A_{i_{m,n}}(\alpha)$ is replaced by $1-\alpha$ if $m = n = i+1$. The following (3) is the expanding form of (2).

$$\boldsymbol{y}H(\alpha) = \begin{bmatrix} y_1 & y_2 & \cdots & y_t \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & -1 \end{bmatrix}$$
$$\begin{bmatrix} 1 & \alpha & \cdots & 0 \\ 0 & 1-\alpha & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha \\ 0 & 0 & \cdots & 1-\alpha \end{bmatrix}. \tag{3}$$

3.3 LSA Penalty

Since AD is irreversible, the patient's cognitive score may fluctuate at adjacent test time points, but overall, it should not fluctuate dramatically. That means the value of Δy_i is not going to be large, i.e., chasing the global temporal smoothness. Due to the use of linear regression, we have

$$y_i - y_{i+1} = \boldsymbol{x}^T \boldsymbol{w_i} - \boldsymbol{x}^T \boldsymbol{w_{i+1}} = \boldsymbol{x}^T (\boldsymbol{w_i} - \boldsymbol{w_{i+1}}),$$

$\boldsymbol{x}$ represents MRI features of instance, $\boldsymbol{w_i}$ is the ith column of W corresponding to task i. We therefore have the following LSA penalty:

$$\|WH\|_1 = \|WH(\alpha)\|_1.$$

where α is a parameter, $\|X\|_1 = \sum_i \sum_j |x_{ij}|$. An alternative formulation would use $\|WH\|_F^2$ in place of $\|WH\|_1$. However, this does not produce a sparse solution which is attractive for high interpretability. Another choice would use infinity norm, but it only penalizes the maximum deviation of cognitive scores over time, and thus ignores the others.

3.4 One Novel MTL Framework

We first introduce the $L_{2,1} - norm$, well known as group Lasso, denoted by $\|W\|_{2,1} = \sum_{i=1}^{p} \|\boldsymbol{w^i}\|_2$, $\boldsymbol{w^i}$ is the ith row vector of W. Group Lasso can select a common set of biomarkers for all tasks. The constraint of group Lasso is, however, too strict to select the features for specific task. We then utilize the $L_1 - norm$, Lasso , to introduce sparsity for model coefficient matrix. The combination of $L1 - norm$ and $L_{2,1} - norm$ is also known as the sparse group Lasso [32], which encourages simultaneous joint feature selection for all tasks and selection of a specific feature set for each task. Finally, LSA penalty is employed to adaptively incorporate the correlation among tasks.

The novel multi-task learning framework with LSA penalty (MTL-LSA) is defined as:

$$\min_W L(W) + \lambda_1 \|W\|_1 + \lambda_2 \|W\|_{2,1} + \lambda_3 \|WH\|_1, \tag{4}$$

where $\lambda_1, \lambda_2, \lambda_3, \alpha$ are all fine-tuned parameters.

4 TWO OPTIMIZATION ALGORITHMS

In this section, we develop two optimization algorithms, the AGM-based and ADMM-based algorithms to solve the objective function of MTL-LSA.

4.1 The AGM-Based Algorithm

Due to the optimal convergence rate for the class of first-order methods, i.e., $O(1/\sqrt{\epsilon})$ iterations for the desired accuracy ϵ, the accelerated gradient method (AGM) [26] has been extensively utilized to solve multi-task learning problems [13] of the following form:

$$\min_W F(W) = f(W) + g(W), \tag{5}$$

where $f(W)$ is convex and smooth, $g(W)$ is convex but nonsmooth.

The AGM is based on two sequences, the approximation point $\{W_i\}$ and the search point $\{S_i\}$. S_i is the affine combination of W_{i-1} and W_i,

$$S_{i+1} = W_i + \alpha_i (W_i - W_{i-1})$$

where α_i is the combination coefficient, which follows the strategy in [3]. The approximation point W_i is computed as

$$W_i = \pi(S_i - \frac{1}{L} F'(S_i)), \tag{6}$$

where $\pi(V)$ is the proximal operator of V, $\frac{1}{L}$ is the stepsize.

4.1.1 Estimation of the Lipschitz Constant. An appropriate step size of $1/L$ is important for the global convergence of the accelerated gradient based algorithms. The stepsize $1/L$ can be estimated with many sophisticated line-search schemes [4] in general. The value of L is updated until satisfying

$$f(W_i) \leqslant f_L(W_i, S_i) \tag{7}$$

$$= f(S_i) + \langle \nabla f(S_i), W_i - S_i \rangle + \frac{L}{2} \|W_i - S_i\|_F^2. \tag{8}$$

However, this updating procedure may incur overhead costs in the computation, especially the case in which the dimension of dataset is not very large, i.e., not several millions[3].

Inspired by [7], to avoid the expensive computation cost of estimating the Lipschitz constant for $f(W)$, we directly compute the smallest Lipschitz constant, as shown in the following lemma.

Lemma 4.1. *Let $X = [X_1, \cdots, X_t], X_i \in R^{n_i \times p}$, $Y = [\boldsymbol{y_1}, \cdots, \boldsymbol{y_t}]$, $\boldsymbol{y_i} \in R^{n_i}$. The best Lipschitz constant L_f of the function $f(W)$ is no larger than σ_X^2, where $\sigma_X = \max\{\sigma_{X_i}\}, i \in \{1, \cdots, t\}$, σ_{X_i} is the largest singular value of X_i.*

Proof. This proof is similar to [7], which however only considers the scenario all tasks have same samples. □

The most pivotal step in the AGM-based algorithm is computing the proximal operator associated with the composite of three nonsmooth penalties (6). For a more detailed form, we rewrite it as

$$\pi(V) = \arg\min_W \frac{1}{2} \|W - V\|_F^2 + \lambda_1 \|W\|_1 + \lambda_2 \|W\|_{2,1} + \lambda_3 \|FW^T\|_1. \tag{9}$$

where $V = S_i - \frac{1}{L_f} f'(S_i), F = H^T$. It is clear that each row of W is decoupled in (9). Thus for obtaining $\boldsymbol{w_i}$, the ith row of matrix W, we need to solve

$$\pi(\boldsymbol{v_i}) = \arg\min_{\boldsymbol{w_i}} \frac{1}{2} \|\boldsymbol{w_i} - \boldsymbol{v_i}\|_2^2 + \lambda_1 \|\boldsymbol{w_i}\|_1 + \lambda_2 \|\boldsymbol{w_i}\|_2 + \lambda_3 \|F\boldsymbol{w_i}\|_1, \tag{10}$$

where $\boldsymbol{v_i}$ is the i-th row of V.

4.1.2 Computing the Proximal Operator. In recent years, the alternating direction method of multipliers (ADMM) [5] has attracted much focus, since it is easy to parallelize distributed convex problems. Now we show how to solve the equation (10), based on ADMM, which is equivalent to the following constrained optimization problem:

$$\min_{\boldsymbol{w}, \boldsymbol{z_1}, \boldsymbol{z_2}} \frac{1}{2} \|\boldsymbol{w} - \boldsymbol{v}\|_2^2 + \lambda_1 \|\boldsymbol{z_1}\|_1 + \lambda_2 \|\boldsymbol{z_1}\|_2 + \lambda_3 \|\boldsymbol{z_2}\|_1 \quad \text{s.t. } \boldsymbol{w} = \boldsymbol{z_1}, F\boldsymbol{w} = \boldsymbol{z_2}, \tag{11}$$

where $\boldsymbol{z_1}, \boldsymbol{z_2}$ are slack variables, also known as auxiliary variables.

It is worth noting that we use only one single auxiliary vector $\boldsymbol{z_1}$ to relax both Lasso penalty and group Lasso penalty to reduce

the computational complexity. But trivial ADMM algorithm needs two auxiliary variables for two nonsmooth penalties.

Eq. (11) can be solved by ADMM. The augmented Lagrangian function is

$$\begin{aligned} L_\rho(\boldsymbol{w}, \boldsymbol{z_1}, \boldsymbol{z_2}, \boldsymbol{y_1}, \boldsymbol{y_2}) &= \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{v}\|_2^2 \\ &+ \lambda_1\|\boldsymbol{z_1}\|_1 + \lambda_2\|\boldsymbol{z_1}\|_2 + \lambda_3\|\boldsymbol{z_2}\|_1 \\ &+ \boldsymbol{y_1}^T(\boldsymbol{w} - \boldsymbol{z_1}) + \frac{\rho}{2}\|\boldsymbol{w} - \boldsymbol{z_1}\|_2^2 \\ &+ \boldsymbol{y_2}^T(F\boldsymbol{w} - \boldsymbol{z_2}) + \frac{\rho}{2}\|F\boldsymbol{w} - \boldsymbol{z_2}\|_2^2. \end{aligned} \tag{12}$$

From the augmented Lagrangian in (12), the update of w at $(k+1)$th iteration is carried out by

$$\begin{aligned} &((1+\rho)I_{t\times t} + \rho F^T F)\boldsymbol{w^{k+1}} \\ &\quad = \boldsymbol{v^k} - \boldsymbol{y_1^k} + \rho\boldsymbol{z_1^k} - F^T\boldsymbol{y_2^k} + \rho F^T\boldsymbol{z_2^k}. \end{aligned} \tag{13}$$

It is obvious the coefficient matrix of $\boldsymbol{w}$ is symmetric positive definite, so Cholesky factorization is suitable to reduce the computational effort of this step.

Updating $\boldsymbol{z_1}$ needs to solve the following problems:

$$\begin{aligned} \boldsymbol{z_1^{k+1}} = \arg\min_{\boldsymbol{z_1}} &\frac{1}{2}\|\boldsymbol{z_1} - (\boldsymbol{w^k} + \frac{\boldsymbol{y_1^k}}{\rho})\|_2^2 \\ &+ \frac{1}{\rho}(\lambda_1\|\boldsymbol{z_1}\|_1 + \lambda_2\|\boldsymbol{z_1}\|_2). \end{aligned} \tag{14}$$

We introduce the following two lemmas to solve (14) efficiently.

Lemma 4.2. *[24] For any $\lambda \geqslant 0, \boldsymbol{v} \neq \boldsymbol{0}$,*

$$\begin{aligned} \pi(\boldsymbol{v}) &= \arg\min_{\boldsymbol{w}} \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{v}\|_2^2 + \lambda\|\boldsymbol{w}\|_2 \\ &= \max\{\|\boldsymbol{v}\|_2 - \lambda, 0\}\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|_2} \end{aligned}$$

Lemma 4.3. *[45][40] For any λ_1, λ_2,*

$$\pi_{Lasso}(\boldsymbol{v}) = \arg\min_{\boldsymbol{w}} \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{v}\|_2^2 + \lambda_1\|\boldsymbol{w}\|_1$$

$$\pi_{GLasso}(\boldsymbol{v}) = \arg\min_{\boldsymbol{w}} \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{v}\|_2^2 + \lambda_2\|\boldsymbol{w}\|_2$$

$$\pi(\boldsymbol{v}) = \arg\min_{\boldsymbol{w}} \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{v}\|_2^2 + \lambda_1\|\boldsymbol{w}\|_1 + \lambda_2\|\boldsymbol{w}\|_2$$

Then the following holds:

$$\pi(\boldsymbol{v}) = \pi_{GLasso}(\pi_{Lasso}(\boldsymbol{v}))$$

Updating $\boldsymbol{z_2}$ needs to compute

$$\boldsymbol{z_2^{k+1}} = \arg\min_{\boldsymbol{z_2}} \frac{1}{2}\|z_2 - (F\boldsymbol{w^k} + \frac{\boldsymbol{y_2^k}}{\rho})\|_2^2 + \frac{\lambda_3}{\rho}\|z_2\|_1. \tag{15}$$

Following standard ADMM dual variable update schema,

$$\boldsymbol{y_1^{k+1}} = \boldsymbol{y_1^k} + \tau\rho(\boldsymbol{w^{k+1}} - \boldsymbol{z_1^{k+1}}) \tag{16}$$

$$\boldsymbol{y_2^{k+1}} = \boldsymbol{y_2^k} + \tau\rho(F\boldsymbol{w^{k+1}} - \boldsymbol{z_2^{k+1}}) \tag{17}$$

where $\tau \in (0, \frac{1+\sqrt{5}}{2})$ according to [11].

We conclude the whole procedure in Algorithm 1 and 2.

Algorithm 1 Solve the Proximal Operator

Input: $\lambda_1, \lambda_2, \lambda_3, \alpha, \rho, \tau$

Output: $\boldsymbol{w}$

Initialization: $\boldsymbol{w_0} = 0, \boldsymbol{z_1} = 0, \boldsymbol{z_2} = 0, \boldsymbol{y_1} = 0, \boldsymbol{y_2} = 0$.

repeat

Update $\boldsymbol{w^{k+1}}$ according to (13).

Update $\boldsymbol{z_1^{k+1}}$ according to (14).

Update $\boldsymbol{z_2^{k+1}}$ according to (15).

Update $\boldsymbol{y_1^{k+1}}$ according to (16).

Update $\boldsymbol{y_2^{k+1}}$ according to (17).

until Stopping criterion is satisfied.

Algorithm 2 The AGM-based Algorithm

Input: $X, Y, \lambda_1, \lambda_2, \lambda_3, \alpha, L_f$

Output: W

Initialization: $W_0 = W_{-1}$, $t_{-1} = 0$, $t_0 = 1$, and $i = 1$

repeat

$\alpha_i = \frac{t_{i-1}-1}{t_i}$

$S_i = W_{i-1} + \alpha_{i-1}(W_{i-1} - W_{i-2})$

compute $W_i = \pi(S_i - \frac{1}{L_f}f'(S_i))$ by Alg. 1

$t_i = \frac{1+\sqrt{1+4t_{i-1}^2}}{2}$

until convergence

4.1.3 Complexity of The AGM-based Algorithm. The convergence rate of AGM is $(O)(1/\sqrt{\epsilon})$iterations for a desired accuracy ϵ[3]. The complexity of every iteration of the AGM-based algorithm consists of two parts. First is computing the gradient of smooth component $f(W)$, the complexity is $O(npt)$. The second is solving the proximal operator (9), which is dominant in every iteration of the AGM-based algorithm.

The computational cost of solving the proximal operator (10) by ADMM, also has two main parts. The computational cost of Cholesky decomposition is $O(t^3)$ and the update of $\boldsymbol{w_i}$, the ith row of W, is $O(t^2)$. It is worth noting that we usually set the maximum iteration number k in ADMM to several thousand to get a enough precise proximal operator. Hence, the computation cost to get the proximal operator of W in (9) is $O(p(t^3 + kt^2))$.

We conclude that the complexity of the AGM-based algorithm is

$$O((1/\sqrt{\epsilon})(npt + p(t^3 + kt^2))) \tag{18}$$

for a desired accuracy ϵ, where k is set to several thousands.

4.1.4 Convergence with Inexact Proximal Operator. The AGM-based algorithm has the great superlinear convergence rate and is suitable for our frameworks, there is, however, a fatal result in the case of high accuracy. Due to the absence of the analytical solution of (9), we can only compute an inexact solution, which often makes the AGM-based algorithm can not hold the optimal convergence rate [16][30], especially in the case in which high accuracy is needed.

In our numerical experiment shown in Section 5, we notice two phenomena. First, when the needed accuracy is relatively high, the AGM-based algorithm gets stuck at some random values very often, even though the stopping criterion of the proximal operator is extremely strict, with huge computational effort. Second, as the value of accuracy ϵ decreases, meaning the model is required to be more accurate, the needed iteration number is much higher than $O(1/\sqrt{\epsilon})$, which is AGM supposed to be, even several orders of magnitude.

It is concluded that the AGM-based algorithm is suitable for low accuracy case with much efficiency, but basically not to high accuracy case. So we design another optimization algorithm based on inexact ADMM [15], which is shown to have same convergence rate as exact strategy [5], to improve the efficiency.

4.2 The ADMM-Based Algorithm

The original equation (4) is equivalent to the following constrained problem:

$$\min_{W,A,B,C} \frac{1}{2}\|XW - Y\|_F^2 + \lambda_1\|A\|_1 + \lambda_2\|A\|_{2,1} + \lambda_3\|B\|_1 \quad \text{s.t. } W = A, WH = B. \tag{19}$$

We emphasize that the efficient ADMM-based algorithm has some similarities to (11) in form, but they are extremely different, on account of the following reasons:

• Eq. (11) is based on the row decoupling property of (9), but (19) can not be decoupled in row, due to the existence of the first term and last term, that couple all columns of W.

• The update of $\boldsymbol{w}$ in (11) is easy to use the Cholesky factorization to reduce the computational complexity, but the update of W in (19) is the main challenge. First of all, different task has different sample data in usual multi-task models, so we need to conduct column decoupling on W, i.e., decouple all tasks to update W. However, the LSA penalty, as well as Group Lasso penalty, couple all tasks to employ the complicated correlation among tasks to improve the overall performance of our proposed models.

Now we show the update of W can be conducted in an efficient way using a suitably linearization method. Based on this method, the following proposed inexact ADMM based algorithm, which is shown to have same convergence rate as exact ADMM [25], will greatly improve the efficiency compared with the AGM based algorithm.

4.2.1 Update W. From the augmented Lagrangian function of (19),

$$\begin{aligned} L_\rho(W,A,B,C,D) &= \frac{1}{2}\|XW - Y\|_F^2 \\ &+ \lambda_1\|A\|_1 + \lambda_2\|A\|_{2,1} + \lambda_3\|B\|_1 \\ &+ Tr(C^T(W - A)) + \frac{\rho}{2}\|W - A\|_F^2 \\ &+ Tr(D^T(WH - B)) + \frac{\rho}{2}\|WH - B\|_F^2. \end{aligned} \tag{20}$$

the update of W at $(k+1)$th iteration is carried out by setting the gradient of W to 0, we have

$$X^TXW + \rho W + \rho WF = X^TY - C + \rho A - \rho DG + \rho BG, \tag{21}$$

where $F = HH^T, G = H^T$.

Conducting the column of decoupling, yields

$$X_i^TX_i\boldsymbol{w_i} + \rho\boldsymbol{w_i} + \rho W\boldsymbol{f_i} = X_i^T\boldsymbol{y_i} - \boldsymbol{c_i} + \rho\boldsymbol{a_i} - \rho D\boldsymbol{g_i} + \rho B\boldsymbol{g_i}. \tag{22}$$

For simple representation, we have

$$L_i\boldsymbol{w_i} = \boldsymbol{q_i}, i \in \{1, \cdots, t\}, \tag{23}$$

$$L_i = X_i^TX_i + \rho(1 + F_{ii})I_{p\times p}, \tag{24}$$

$$\boldsymbol{q_i} = X_i^T\boldsymbol{y_i} - \boldsymbol{c_i} + \rho\boldsymbol{a_i} - \rho D\boldsymbol{g_i} + \rho B\boldsymbol{g_i} - \rho\sum_{k=1,k\neq i}^{t}\boldsymbol{w_k}F_{ki}. \tag{25}$$

Clearly, $F_{ii} = \boldsymbol{h^i}\boldsymbol{h^i}^T \geqslant 0$, so $L_i, i \in \{1, \cdots, t\}$ is symmetric positive definite, which Cholesky factorization is applicable for, resulting in efficient updating of W.

The update of other variables A, B, C, D is similar to the procedure in Algorithm 1, so we only show the result rather than many details.

$$A^{k+1} = \arg\min_A \frac{1}{2}\|A - (W + \frac{D}{\rho})\|_F^2 + \frac{1}{\rho}(\lambda_1\|A\|_1 + \lambda_2\|A\|_{2,1}). \tag{26}$$

$$B^{k+1} = \arg\min_B \frac{1}{2}\|B - (WH + \frac{E}{\rho})\|_F^2 + \frac{\lambda_3}{\rho}\|B\|_1. \tag{27}$$

$$C^{k+1} = C^{k+1} + \tau\rho(W^{k+1} - A^{k+1}). \tag{28}$$

$$D^{k+1} = D^{k+1} + \tau\rho(W^{k+1}H - B^{k+1}). \tag{29}$$

The whole procedure of the ADMM-based algorithm is similar to Alg. 1, so we do not show the pseudo code.

4.2.2 Complexity of The ADMM-based Algorithm. The convergence rate of ADMM is $O(1/\epsilon)$ iterations for a desired accuracy ϵ [29][18] [17]. The time complexity mainly consists of two parts. First is t times Cholesky factorization of $L_i \in R^{p\times p}$, the total computational cost is $O(tp^3)$. Second is the computational complexity of every iteration, dominated by the update of W (25), is approximately equal to $O(tp(p+n))$. So the whole computational cost is $O(tp^3 + \frac{1}{\epsilon}(tp(p+n)))$.

4.3 Compare the Complexity

In the context of AD data sets we use in Section 5, the maximum number of task instances n is almost one thousand, the MRI feature number p is 314, the total task number t is 12, and k is several thousand in our experimental setting, so we conclude that computational complexity of different algorithms in Table 1.

Table 1: The Comparison of Convergence Rate and Complexity of Two Proposed Algorithms for A Desired Accuracy ϵ.

Algorithm	AGM-based	ADMM-based
Convergence Rate	$O(1/\sqrt{\epsilon})$	$O(1/\epsilon)$
General Complexity	$O(\frac{1}{\sqrt{\epsilon}}(npt + p(t^3 + kt^2)))$	$O(tp^3 + \frac{1}{\epsilon}(tp(p+n)))$
Complexity in AD	$O(\frac{1}{\sqrt{\epsilon}}pkt^2)$	$O(tp^3 + \frac{1}{\epsilon}tpn)$

According to Table 1, if ϵ is relatively large, $1/\epsilon$ is much less than the constant k, the computational cost of the AGM-based

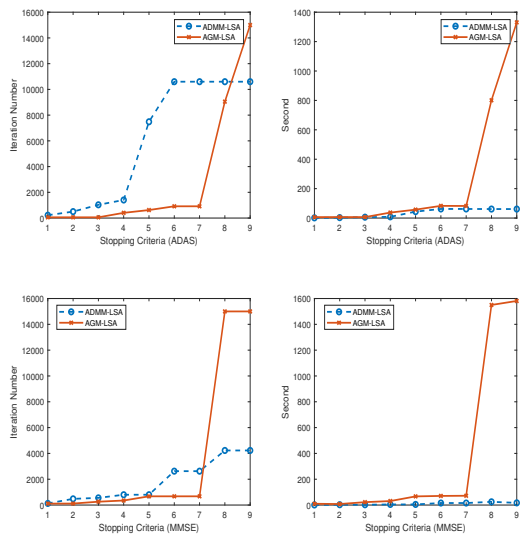

Figure 2: Comparison of the AGM-based and ADMM-based algorithms on ADAS dataset, in terms of the iteration number (left plot) and the computational cost (right plot). The x-axis indexes the stopping criterion from 10^2 to 10^{-6}.

algorithm and the ADMM-based algorithm is similar. If ϵ is small, the computational cost of AGM-based algorithm would be smaller than ADMM-based algorithm, just from theoretical point of view. But as we discussed before, the AGM-based method is not suitable for the case in which high accuracy is needed, due to the absence of the explicit solution of proximal operator (9).

We conclude that if only low accuracy is needed, both the AGM-based and ADMM-based algorithms are suitable with high efficiency and similar computational complexity. If high accuracy is necessary, the ADMM-based algorithm is a much better option.

5 EXPERIMENTAL RESULT

5.1 The Latest Dataset from ADNI

The AD dataset from ADNI project is a longitudinal study in which various measurement data are collected repeatedly from selected subjects at intervals of 6 months or 1 year. The date when the patient was screened in the hospital for the first time is called the baseline, and the time point of follow-up is represented by the time when the baseline started. For instance, we use the notation "M06" to denote the time point half year after the first visit. Currently the latest ADNI has up to 120 months' follow-up data available for some patients. The two measurements we use in this paper are MMSE and ADAS. The data preprocessing we execute includes: ❶ Exclude patients without baseline MRI records. ❷ Delete subjects with failed quality control of MRI image. ❸ Complete missing entries of features using the average value. We finally get a total of 314 features. There are 12 time points in our experiment dataset.

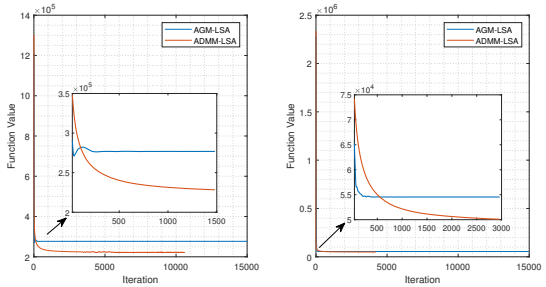

Figure 3: Convergence rate comparison between the AGM-based and ADMM-based algorithms. The left plot is conducted on ADAS dataset, and the right plot is on MMSE dataset.

5.2 Comparison of Two Algorithms

We empirically compare the AGM-based and the ADMM-based algorithms, using ADNI dataset. The optimization algorithms are implemented in Matlab.

Because the training ratio is 0.9 in the setting of Subsection 5.3, we stochastically select 90% of two data sets to conduct the comparison of the proposed algorithm. Due to the limitation of paper space, we only show the comparison result with randomly selected parameters $\lambda_1 = 1, \lambda_2 = 100, \lambda_3 = 50, \alpha = 0.1$ for MMSE dataset, and $\lambda_1 = 0.1, \lambda_2 = 100, \lambda_3 = 100, \alpha = 0.2$ for ADAS dataset. For other parameter settings, we observe similar trends in our experiments. The algorithm is terminated when the change of objective function values in two successive steps is not more than the stopping criterion. From Fig. 2, we find in the case of low accuracy, i.e., the stopping criterion is roughly in $\{10^2, \cdots, 10^{-3}\}$, the AGM-based algorithm has much faster convergence rate than the ADMM-based algorithm. But the time cost is more expensive than the ADMM-based algorithm, because every iteration in the AGM-based algorithm needs to compute the proximal operator of three nonsmooth penalties, which is time-consuming. In the case of high accuracy, means stopping criterion is roughly in $\{10^{-4}, 10^{-5}, 10^{-6}\}$, the ADMM-based algorithm works well, and the AGM-based algorithm has some severe problems. Refer to 3, it is clear the superlinear convergence rate has gone when the iteration number is big (≥ 200), the function values generated by the AGM-based algorithm get stuck at some random values. The numbers of iterations of the AGM-based algorithm all get to the maximal number, which we set to 15000.

5.3 Empirical Evaluation

In this section, we use 90% of the data as training data, repeat 10 trials, five-fold cross validation is done to select the regularization parameters. The data is z-scored.

5.3.1 Evaluation Metrics. For quantitative performance evaluations, we use the metric of Root Mean Squared Error (rMSE) between the predicted cognitive scores and the target cognitive scores for the task-specific regression performance measures. Moreover, to evaluate the overall performance on all tasks, we exploit the normalized mean squared error (nMSE) as usually used in the multi-task

Table 2: Comparison of models with different task numbers in terms of nMSE and wR. 90% of data is used as training data. The shown data in this table is mean value ± standard deviation.

Data	Task	Metric	DMTL	RMTL	TGL	cFSGL	MNorm	RMTFL	NC	FL	MTL-LSA
ADAS	4	nMSE	0.50±0.03	0.56±0.04	0.47±0.02	**0.46±0.03**	**0.46±0.02**	0.56±0.04	**0.46±0.03**	0.52±0.03	**0.46±0.03**
		wR	0.71±0.02	0.68±0.02	0.73±0.01	**0.75±0.01**	0.74±0.02	0.68±0.03	**0.75±0.02**	0.69±0.03	0.74±0.02
	8	nMSE	0.55±0.04	0.62±0.04	0.50±0.02	0.49±0.02	0.52±0.03	0.62±0.04	0.49±0.02	0.65±0.04	**0.46±0.02**
		wR	0.67±0.02	0.65±0.02	0.71±0.03	0.72±0.02	0.71±0.03	0.65±0.02	**0.73±0.01**	0.60±0.03	**0.73±0.01**
	12	nMSE	0.58±0.02	0.68±0.03	0.51±0.03	0.50±0.02	0.59±0.02	0.68±0.04	0.54±0.03	0.84±0.05	**0.48±0.02**
		wR	0.68±0.01	0.63±0.03	0.71±0.02	0.73±0.02	0.67±0.02	0.63±0.03	0.71±0.03	0.45±0.04	**0.74±0.02**
MMSE	4	nMSE	0.65±0.07	0.75±0.11	0.58±0.07	0.58±0.06	0.69±0.10	0.75±0.11	0.61±0.05	0.72±0.06	**0.57±0.06**
		wR	0.61±0.03	0.57±0.66	**0.67±0.26**	**0.67±0.03**	0.61±0.04	0.57±0.07	0.66±0.02	0.55±0.05	**0.67±0.03**
	8	nMSE	0.77±0.08	0.75±0.05	0.65±0.07	0.68±0.09	0.67±0.03	0.75±0.05	0.76±0.07	0.87±0.07	**0.62±0.04**
		wR	0.59±0.03	0.59±0.02	0.64±0.03	**0.65±0.02**	0.61±0.02	0.59±0.03	0.61±0.01	0.43±0.08	**0.65±0.02**
	12	nMSE	0.74±0.08	0.82±0.07	0.67±0.07	0.65±0.04	0.72±0.05	0.82±0.07	0.81±0.04	0.92±0.01	**0.61±0.04**
		wR	0.57±0.02	0.58±0.02	0.65±0.01	0.64±0.02	0.60±0.02	0.58±0.03	0.59±0.01	0.45±0.03	**0.66±0.01**

learning literature [36, 41, 47] and weighted R-value (wR) as employed in the medical literature [33]. The lower value of nMSE and rMSE means better performance. wR is the opposite.

5.3.2 Comparative Models. We comprehensively compare our proposed methods with multiple popular state-of-the-art related methods. We thank to [43] to provide the implementation code of several methods. All comparative models are include: Dirty model for multi-task learning (DMTL)[20], Robust multi-task learning (RMTL) [8], Temporal group lasso regularization (TGL) [46], Convex fused group Lasso (cFSGL) [44], Robust multi-task feature learning (RMTFL) [13], Learning Incoherent Sparse and Low-Rank Patterns (Mnorm), Calibrated multi-task learning based on non-convex relaxation (NC-CMTL, NC) [28], Fused Laplacian Sparse Group Lasso (FL) [25]. The tuning range of all parameters is $10^i, i \in \{-2, -1, \cdots, 3\}$. For our MTL-LSA, $\alpha \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2\}$. We stop the algorithm if the relative change of the objective value in two consecutive iterations is not more than 10^{-4}.

5.3.3 Experiment Analysis. We study three scenarios with different task numbers from baseline (M00) to M24, M72 and M120, respectively. As shown in Table 2, MTL-LSA has the best performance regardless of the number of tasks. TGL and cFSGL have the second best performance. They are designed specifically for modeling AD progression. TGL and cFSGL both utilize the temporal smoothness assumption, but they are not good as our proposed model. It means the effectiveness of LSA, also the full use of the information accumulated in the process of disease development is a better choice. In addition, DMTL and NC also have excellent performances, which not only indicates the noise has a great influence on the used dataset, but the data distribution of dataset is unbalanced. Note that, RMTL and RMTFL performance is not very good, which means the impact of outlier task is not great. Although FL is designed for modeling AD progression, does not have great performance. The possible reason is the selected bandwidth constant σ is not proper.

Obviously, our model works better on the ADAS data set than on the MMSE dataset. Running on ADAS dataset, the values of nMSE and wR of our model are approximately $0.45 \sim 0.51$ and $0.72 \sim 0.74$, while the values of nMSE and wR are approximately $0.57 \sim 0.62$ and $0.65 \sim 0.67$ on MMSE dataset. Other comparative models have same experimental results. The conclusion is ADAS maybe a better clinical standard for AD. We emphasize that the average α of the experiments conducted on ADAS dataset and MMSE dataset, is 0.045 and 0.062. It demonstrates how the previous states influences the current diagnosis. We also study the prediction performance at every single time point, shown in Fig. 4. It is obvious the prediction performance of MTL-LSA is basically optimal at all time points, which shows the superiority of our proposed model. Note that, the performance changes of all models have similar trends at all time points. The results on the ADAS dataset show that the prediction performances of all models are worse in the later stage. For example, the rMSE of MTL-LSA is approximately 6.5 at M00, but has reached the highest value of 11 at M120. The same is true for other models, such as MNorm with an rMSE of almost 19 at M120. The may reason is there are fewer samples in the later period.

The results on MMSE dataset show the prediction ability of all models at a single time point fluctuates greatly, and even there is no clear trend in the later stage. The possible reason is MMSE dataset is more disturbed by noise. Our proposed model still has the best prediction results for every single time point. This shows that the adaptive LSA we propose to capture and exploit complex relationships between multiple tasks enforces our model to have not only the best overall performance, but also the best performance at every single task.

5.4 The Identification of Longitudinal MRI Biomarkers

One of the strengths of MTL-LSA is to facilitate the identification of temporal patterns of biomarkers. We use the method of longitudinal stability selection [45] to study the MRI biomarkers identified by our formulations. Generally speaking, the stability selection procedure consists in running our framework MTL-LSA to numerous random subsets of data and computing the frequency of each feature is selected (corresponding model weight is greater than a threshold value) for each cognitive score and time point across the runs. For more details, see [45]. The calculated frequency vector is called a stability vector here.

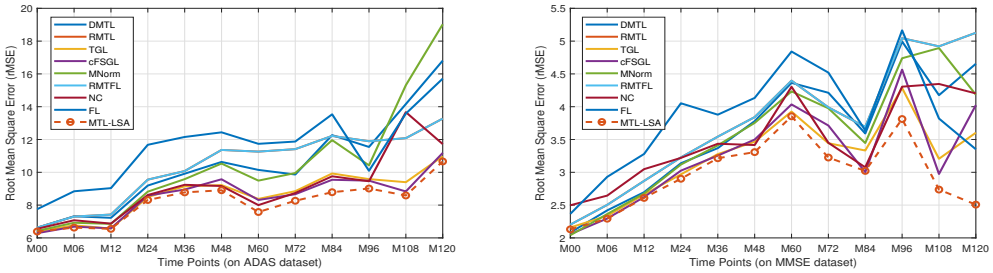

Figure 4: Comparison of rMSE at each time point. The left plot is conducted on ADAS dataset and the right plot on MMSE.

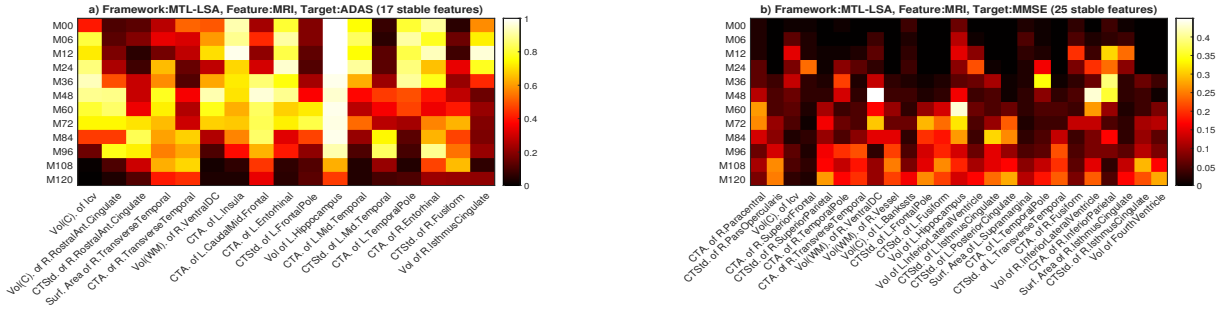

Figure 5: The stability vector of stable MRI features using MTL-LSA. The selection is conducted on ADAS and MMSE dataset.

5.4.1 Temporal Pattern of Stable Biomarkers. The datasets used in existing paper are not the newest, and later task patients are rarely, so many researchers usually consider the stable MRI features of the first four or five time points[44][45][25], but we have adopted the latest AD data, and the experimental result in Subsection 5.3 has shown that although the samples of M108 and M120 are very few, our model still has good performance. So we study the temporal pattern of stable MRI features at all time points, from M00 to M120.

Our stability selection is set to select the four most stable biological features at each time point, and the final experimental results are shown in Fig. 5. First of all, we find there are no extremely stable features at M108 and M120, that means the impact of lacking samples can not be ignored easily. We observe that the white matter volume of left hippocampus and the cortical thickness average of left and right entorhinal are the most outstanding stable features for all time points. The cortical thickness average of left insula and cortical thickness average of left temporal pole provide significant information in the first five time points. Several biomarkers including white matter volume of right parahippocampel, cortical thickness standard of right rostral anterior cingulate, and surface area of transverse temporal, provide useful information only in later time points. On the contrary, some biomarkers have a large stability score during first five time points, such as cortical thickness average of left temporal pole and white volume of right isthmus cingulate.

On MMSE dataset, we obtain very different temporal patterns from ADAS dataset. The number of stability vectors of MRI features on MMSE dataset is 25, much larger than the number on ADAS dataset. And no feature is very stable as on ADAS dataset. The white matter volume of left hippocampus is still the most stable features on MMSE datset, same as the result on ADAS dataset. Sevral features only provide significant information for the several middle time points, from M24 to M60, including volume of right ventral IDC and white matter volume of right inferior parietal. The different temporal patterns of the biomarkers of these two scores also indicate that restricting the model for predicting these two scores to share a common set of features, as done in previous work, may lead to sub-optimal performance.

6 CONCLUSION

We proposed one novel longitudinal stability adjustment (LSA) method to adaptively capture the complex task correlation. A new framework MTL-LSA, based on LSA, is presented to utilize the accumulative disease progression information, for modeling disease progression. Two efficient optimization algorithms, based on AGM and ADMM, are designed to tackle the associated difficult optimization problem. The experimental results conducted on the latest ADNI dataset we have preprocessed demonstrate our model is superior to comparative state-of-the-art models.

LSA is a general way to capture the global temporal correlation among tasks, we plan to study the feasibility of applying it to other disease cases, such as Parkinson's disease and diabetes. In addition, although the current researches illustrate the power of MTL models, they only consider one type of cognitive score at one time. We expect to expand our framework to consider different kinds of cognitive scores simultaneously.

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